Bank Customer Subscription Prediction & Customer Segmentation System

Course: DSA0402 – Fundamentals of Data Science
Integrated Assignment (IET)
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📌 Project Overview

This project implements a complete, quality-audited Data Science and Machine Learning solution for a retail banking institution:

1. **Primary Pre-Contact Supervised Classification:** Predict whether a customer will subscribe to a bank term deposit (y = yes/no) using only pre-contact demographic, financial, and historical campaign attributes (eliminating temporal data leakage by excluding post-contact call duration).
2. **Unsupervised Customer Segmentation:** Segment customers into distinct behavioral clusters ($K = 4$) using K-means clustering to inform targeted marketing strategies.

📊 Dataset & Source

- **Dataset:** UCI Bank Marketing Dataset (bank-full.csv)
- **Source:** UCI Machine Learning Repository
- **Reference:** Moro et al., 2014. *A Data-Driven Approach to Predict the Success of Bank Telemarketing*, Decision Support Systems.
- **Records:** 45,211 rows (39,922 non-subscribers, 5,289 subscribers)
- **Attributes:** 17 raw features -> 48 encoded features
- **Target Variable:** y (Term deposit subscription status: yes / no)

📁 Project Structure

IDE-Assessment/

├── Instructions.txt # Master execution specification

├── Jaswanth - IET ASSIGNMENT DSA0402 FODS.docx # Assignment rubric & guidelines

├── README.md # Project documentation

```
├── requirements.txt            # Python dependencies
├── run_assignment.bat          # Automated batch execution script
│
├── data/
│   ├── raw/                    # Raw UCI dataset (bank-full.csv)
│   └── processed/              # Encoded and transformed dataset (bank_processed.csv)
│
├── src/
│   └── bank_customer_analysis.py    # Quality-audited Python pipeline & CLI menu interface
│
├── outputs/
│   ├── tables/                 # Summary CSV tables (EDA, Stats, Clusters, Audits)
│   ├── visualizations/         # High-resolution PNG graphs
│   ├── models/                 # Trained machine learning model artifacts (.pkl)
│   └── metrics/                # Model comparison and confusion matrices
│
└── report/
    ├── DSA0402_IET_Assignment_Report.md # Main comprehensive assignment technical report
    └── individual_contribution.md    # Individual contribution report for Naga Jaswanth B.
```

🛠️ Installation & Setup

1. **Navigate to Project Directory:**

cd IDE-Assessment

2. **Install Required Python Dependencies:**

pip install -r requirements.txt

🚀 How to Run

Option 1: Automated Non-Interactive Execution (Batch Mode)

Run the batch script to execute all data loading, preprocessing, EDA, statistical analysis, primary/secondary model training, clustering, and evidence table export:

run_assignment.bat

or via terminal:

python src/bank_customer_analysis.py --batch

Option 2: Interactive Menu-Driven Interface

Run the interactive CLI menu to explore specific phases of the data science workflow:

python src/bank_customer_analysis.py

🤖 Models & Methodology

Primary Pre-Contact Classification (Deployable, Temporal-Leakage-Free)

1. **k-Nearest Neighbours (kNN):** Distance-weighted classification ($k = 7$, StandardScaler).
2. **Decision Tree / CART:** Gini impurity classifier (max_depth=8, class_weight='balanced').
3. **Logistic Regression:** L2 regularized linear model ($C = 1.0$, class_weight='balanced').

Unsupervised Clustering

- **K-Means Clustering:** Applied on scaled pre-contact features ($K = 4$). Optimal K selected via **Elbow Method** and **Silhouette Analysis** diagnostics; visualized using 2D Principal Component Analysis (PCA).

📈 Key Results Summary

- **Dataset Size:** 45,211 instances (88.30% non-subscribers, 11.70% subscribers).
- **Primary Pre-Contact Model Performance ($n_{test} = 9{,}043$):**
    - **Decision Tree / CART** achieved the best deployable performance with **83.24% Accuracy**, **55.95% Recall**, and the highest **F1-Score (0.4385)**, capturing 592 true subscribers.

    - **Logistic Regression:** 75.54% Accuracy, 62.29% Recall, 0.3734 F1-Score.
    - **kNN:** 88.85% Accuracy, 24.20% Recall, 0.3368 F1-Score.
- **Statistical Inference:** 95% Confidence Interval for Population Mean Customer Balance is **[1,334.21, 1,390.34]** currency units.
- **Customer Clusters (K=4):**
    - **Cluster 0 (High Potential / Prior Engaged):** 18.75% Subscription Rate (12.81% prior success)
    - **Cluster 1 (Mass Housing Debt):** 6.76% Subscription Rate (100% housing loans)
    - **Cluster 2 (Mature Higher-Balance):** 15.06% Subscription Rate (Mean balance 1,635.46 units)
    - **Cluster 3 (Overdrawn / High Risk):** 6.38% Subscription Rate (Mean balance -137.62 units)

📜 Reproducibility & Integrity

- Fixed random seeds (random_state=42) across train-test splits, classification models, PCA, and K-means clustering.
- All numbers, confusion matrices, and tables reported in report/DSA0402_IET_Assignment_Report.md stem directly from actual executed output files.